AI CAREER PULSE: DECODING THE JOB MARKET

Project Report

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1. Introduction

1.1 Project Overview

AI Career Pulse: Decoding the Job Market is a data analytics project developed to provide a clearer understanding of the rapidly changing Artificial Intelligence employment landscape. The project focuses on examining important factors such as job demand, salary levels, job roles, required skills, experience requirements, industries, company size, and geographical opportunities.

The analysis is based on an AI job-market dataset containing approximately 1,500 records and 25 attributes. The collected information was examined, checked for data-quality issues, prepared for analysis, and then transformed into meaningful visualizations using Tableau. The project presents these findings through interactive dashboards and a Tableau story, allowing users to explore different aspects of the AI employment market and compare career opportunities across multiple dimensions.

The overall purpose of the project is to convert raw job-market information into simple and understandable insights that can support students, freshers, and aspiring AI professionals while exploring possible career paths and understanding the skills and qualifications expected by employers.

The following tools and technologies were used throughout the project:

Tool / Technology	Purpose
Tableau Desktop / Tableau Public	Creating visualizations, interactive dashboards, stories, analysis, and publishing
CSV / Excel	Storing and preparing the AI job-market dataset for analysis
Google Drive	Storing and sharing the raw job-market dataset
Microsoft Word	Preparing, organizing, and documenting the complete project report

1.2 Objectives

• Examine AI employment data to identify important patterns in job availability, salary ranges, experience requirements, skills, industries, and locations.

• Determine which AI job categories and individual job roles show stronger demand and higher compensation levels within the available dataset.

• Analyse how factors such as company size, industry, experience level, and location are associated with AI employment opportunities and salary distribution.

• Develop an interactive Tableau dashboard that allows users to explore job-market information through visual comparisons, KPIs, charts, and filters.

• Present complex job-market data in a simple visual format so that students and freshers can compare different AI career options more easily.

• Identify changes in AI hiring activity over time and understand whether the number of available job opportunities is increasing or decreasing.

• Highlight highly demanded skills, promising job roles, salary patterns, and other market indicators that may help aspiring professionals understand current industry expectations.

• Convert the findings from the analysis into meaningful career-oriented observations without making individual employment guarantees or predictions.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Project Initialization and Planning Phase

Date	20 Aug 2026
Project Name	AI Career Pulse : Decoding the Job Market

Define Problem Statements (Customer Problem Statement Template):

This Customer Problem Statement template helps you understand the main issues your customers face and focus on what really matters to them. A clear problem statement can help you and your team come up with better solutions to address these challenges. It also allows you to put yourself in the customer's shoes and better understand how they see and experience your product or service.

PROBLEM STATEMENTS (PS)	CUSTOMER PROBLEM STATEMENT
PS 1	
I am	a student or aspiring professional interested in building a career in Artificial Intelligence.
I'm trying to	understand the available AI career paths and the skills required to enter the job market
But	I find it difficult to identify which AI roles are most suitable for my interests and which skills employers currently demand
Because	AI job requirements, technologies, and career opportunities are changing rapidly, making reliable career information difficult to compare
Which makes me feel	uncertain and overwhelmed about choosing the right career path and preparing myself for future opportunities
PS 2	
I am	a student or fresher exploring different career opportunities in the AI industry
I'm trying to	compare AI-related jobs based on salary, location, experience requirements, and required skills
But	the available job information is scattered across different sources and is difficult to analyze in one place
Because	there is no simple and organized way to understand current AI job market trends and compare different career opportunities

Which makes me feel	confused about which AI career option offers the best opportunities for my skills and future growth

2.2. Project Proposal (Proposed Solution)

Project Initialization and Planning Phase

Date	15 March 2024
Project Title	AI Career Pulse : Decoding the Job Market

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	**Project Overview**
Objective	To analyze the AI job market and create an interactive Tableau dashboard that helps students and freshers understand job roles, salaries, required skills, experience levels, and career opportunities.
Scope	The project will analyze an existing AI job-market dataset using Tableau. It will cover job roles, salaries, skills, experience, education requirements, locations, industries, company size, and other relevant job-market factors.
Problem Statement	**Problem Statement**
Description	Students and freshers interested in AI careers often find it difficult to understand which jobs are in demand, what skills employers expect, and how different career opportunities compare. The information is often scattered and difficult to analyze.
Impact	Solving this problem will help students and aspiring professionals better understand the AI job market and make more informed decisions about their career paths, skills, and future opportunities.
Proposed Solution	**Proposed Solution**
Approach	The dataset will first be cleaned and prepared for analysis. Tableau will then be used to create KPIs, charts, filters, and interactive dashboards to identify patterns in job demand, salaries, skills, experience, and locations.
Key Features	The proposed solution will feature an interactive Tableau dashboard that provides insights into AI job demand, salary trends, popular job roles, in-demand skills, and experience requirements. It will also allow users to explore job opportunities across different locations and industries using interactive filters and visual summaries. Key findings and career-focused recommendations will help students and freshers better understand the AI job market and make informed career decisions.

Section	**Details**
Overview	A Tableau-based analysis of the AI job market designed to help students and freshers understand career opportunities and industry trends.
Objective	To explore AI job demand, salary trends, required skills, and available career opportunities.
Scope	Covers AI job roles, salaries, skills, experience levels, locations, industries, and other important job-market trends.
Problem	Students and freshers often find it difficult to compare AI career roles, required skills, salaries, and available opportunities.
Impact	Helps users make better decisions regarding their career path and the skills they need to develop.
Solution	Analyze AI job-market data and present the findings through an interactive Tableau dashboard.
Features	Interactive KPIs, filters, job and salary comparisons, skill analysis, experience insights, and career-focused recommendations.

2.3. Initial Project Planning

Date	20 August 2026
Project Name	AI-Career-Pulse-Decoding-The-Job-Market

Sprint	**Functional Requirement (Epic)**	**User Story Number**	**User Story / Task**	**Story Points**	**Priority**	**Sprint Start Date**	**Sprint End Date (Planned)**
Sprint-1	Project & Data Understanding	USN-1	I can recognize the main questions and goals involved in analyzing the AI job market.	2	High	20 aug 2026	21 aug 2026
Sprint-1	Data	USN-2	I can gather	3	High	20 aug	21 aug

	Collection		the required AI job-market dataset and understand the available data.			2026	2026
Sprint-1	Data Preparation	USN-3	I can clean and organize the dataset so that it is suitable for further analysis.	3	High	23 aug 2026	24 aug 2026
Sprint-1	Exploratory Analysis	USN-4	As a user, I can register for the application through Gmail	3	High	23 aug 2026	24 aug 2026
Sprint-2	Tableau Visualization	USN-5	I can develop Tableau dashboards and visualizations to present important insights from the AI job-market data.	5	High	26 aug 2026	27 aug 2026
Sprint-2	Dashboard Development	USN-6	I can explore and analyze job demand, salary trends, experience levels, skills, and geographical locations through an interactive dashboard.	5	High	27 aug 2026	28 aug 2026
Sprint-3	Dashboard Interaction	USN-7	I can filter and compare different dimensions such as job roles, locations, experience, and skills.	3	Medium	27 aug 2026	28 aug 2026

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

Data Collection and Preprocessing Phase

Date	15 March 2024
Project Title	AI Career Pulse : Decoding The Job Market

Section	Description
Project Summary	The initiative, titled AI Career Pulse: Deciphering the Employment Market, leverages Tableau to evaluate market data. It assists fresh graduates and students in understanding job demand, compensation trends, skill requirements, work experience levels, geographic locations, industries, and overall career prospects.
Data Strategy	The project utilizes the provided AI job-market dataset as the foundational raw source. This data is designated for exploratory analysis, rigorous quality assessment, preprocessing steps, and subsequent visualization in Tableau.
Raw Data Sources Identified	One raw source file has been pinpointed and utilized: the designated AI job-market dataset encompassing 1,500 rows and 25 attributes. Because a formal source URL or repository link was omitted from the project source materials, none has been fabricated.

Source	Name	Description	Format	Size
Dataset 1	AI Job-Market Dataset.	A raw dataset with records about the AI job market, including details on job titles, salaries, required skills, work experience, locations, hiring demand, and other job market trends.	CSV (later converted into xslx)	1500 records / 25 fields
Location/URL : https://docs.google.com/spreadsheets/d/11Cy3CC9XjHK8l3GUmMErYCZuvOuMTQlU/edit?usp=drive_link&ouid=113542697260687224051&rtpof=true&sd=true				

3.2. Data Quality Report

Prior to conducting analysis in Tableau, the AI job-market dataset underwent a quality review during the initial phase. The evaluation covered 1,500 records.

Source of Data	Identified Quality Issue	Severity Level	Proposed Resolution
AI Job-Market Dataset	Absent values: Across 25 fields, no cells were found to be empty.	Low	No imputation or treatment of missing values is required.
AI Job-Market Dataset	Duplicate entries: No duplicate rows or duplicate Job ID values were found.	Low	No duplicate records require deletion.
AI Job-Market Dataset	Experience mismatch: A discrepancy exists between experience_level and years_of_experience in 1,099 of 1,500 records.	High	Derive and standardize experience level from years of experience using established brackets: Entry 0–2, Mid 3–5, Senior 6–9, Lead 10+.
AI Job-Market Dataset	Salary bound inconsistencies: In 288 instances (19.2%), annual_salary_usd exceeds salary_max_usd.	Moderate	Verify the original definitions and resolve discrepancies according to the source logic.
AI Job-Market Dataset	Salary bracket alignment: 30 records conflict with their designated salary-tier descriptions.	Moderate	Establish uniform salary-tier boundaries and recompute salary tiers using annual salary.
AI Job-Market Dataset	Range verification: Demand score 68–98, benefits score 6.0–9.8, posting month 1–12, posting year 2025–2026; no values breached the ranges.	Low	No corrective action is needed for range validation.

Data Quality Execution Steps

- Missing data: Examine all 25 fields for empty entries. No missing data was detected.
- Duplicate records: Inspect all fields for duplicates. No redundant records were uncovered.
- Experience parameters: Verify alignment between experience levels and years of experience and update experience_level where necessary.
- Salary attributes: Verify annual salary values against minimum and maximum salary thresholds.
- Salary classification: Recompute and standardize salary tiers using annual compensation.
- Final verification: Re-examine all fields after revisions to confirm that the dataset is prepared for Tableau.

3.3. Data Exploration and Preprocessing

Date	15 March 2024
Project Title	AI Career Pulse : Decoding the Job Market

Section	Work Completed / Findings
Data Overview	The dataset consists of approximately 1,500 AI job-market records and 25 variables in CSV format, (which I later converted into xlsx format). It includes details related to job roles, categories, experience, education, salaries, locations, remote work, company size, industries, skills, demand, growth, benefits, and other job-related indicators.
Data Cleaning	The dataset was checked for missing values and duplicate records. No significant missing values or duplicate job entries were identified.
Data Transformation	The existing categorical and numerical fields were retained in their original structure for Tableau analysis. Additional calculated fields and measures can be created later depending on the visualization requirements.
Data Type Conversion	All data types were reviewed to ensure accuracy. Numerical columns are stored as numeric values, while categorical columns are maintained as text. Year and month-related fields can be used for time-based analysis where required.
Column Splitting and Merging	No major column splitting or merging was required because the dataset already contains separate fields for job title, category, city, country, experience, salary, industry, and other relevant dimensions.
Data Modeling	The dataset is structured as a single flat table; therefore, no table relationships are required. The available fields can be directly used in Tableau for dimensions, measures, filters, and calculated fields.
Save Processed Data	A cleaned and validated version of the dataset was prepared and saved for the next stage of analysis and dashboard development in Tableau.

4. Data Visualization

4.1. Framing Business Questions

Business Question and Visualization Report

Date	27 August 2026

The AI job market is changing quickly, with differences in job demand, salaries, industries, company sizes, job roles, and experience requirements. This project uses Tableau to study these patterns and present them through clear and interactive visualizations. The dashboards help users understand salary trends, popular job roles, industry opportunities, differences between organizations, and changes in hiring over time. These insights can help job seekers make better and more informed career decisions.

Business Inquiry / Objective	Visual Medium
Job Category Vs Annual Salary	Treemap Diagram
Job Title Vs Annual Salary	Vertical Bar Graph
Company Size Vs Salary	Horizontal Bar Graph
Key Performance Indicator	KPI Metric Card
Job title Vs Job count	Horizontal Bar Graph
Industry Vs Salary	Segmented Bar Graph
No. of Job Posted by year	Chronological Bar Chart
Top 10 Highest Paying Job Titles	Horizontal Barchart

4.2. Developing Visualizations

The visualization development phase translated the framed business questions into Tableau charts. The selected visual media included treemap diagrams, vertical and horizontal bar charts, KPI metric cards, segmented/stacked bar charts. These visualizations were selected to compare salary, job count, company size, industry, and hiring trends.

Business Questions and Visualisation :

Eight business questions were established to structure the Tableau analysis, with each question linked to an appropriate visualisation for highlighting a particular aspect of the AI job market.

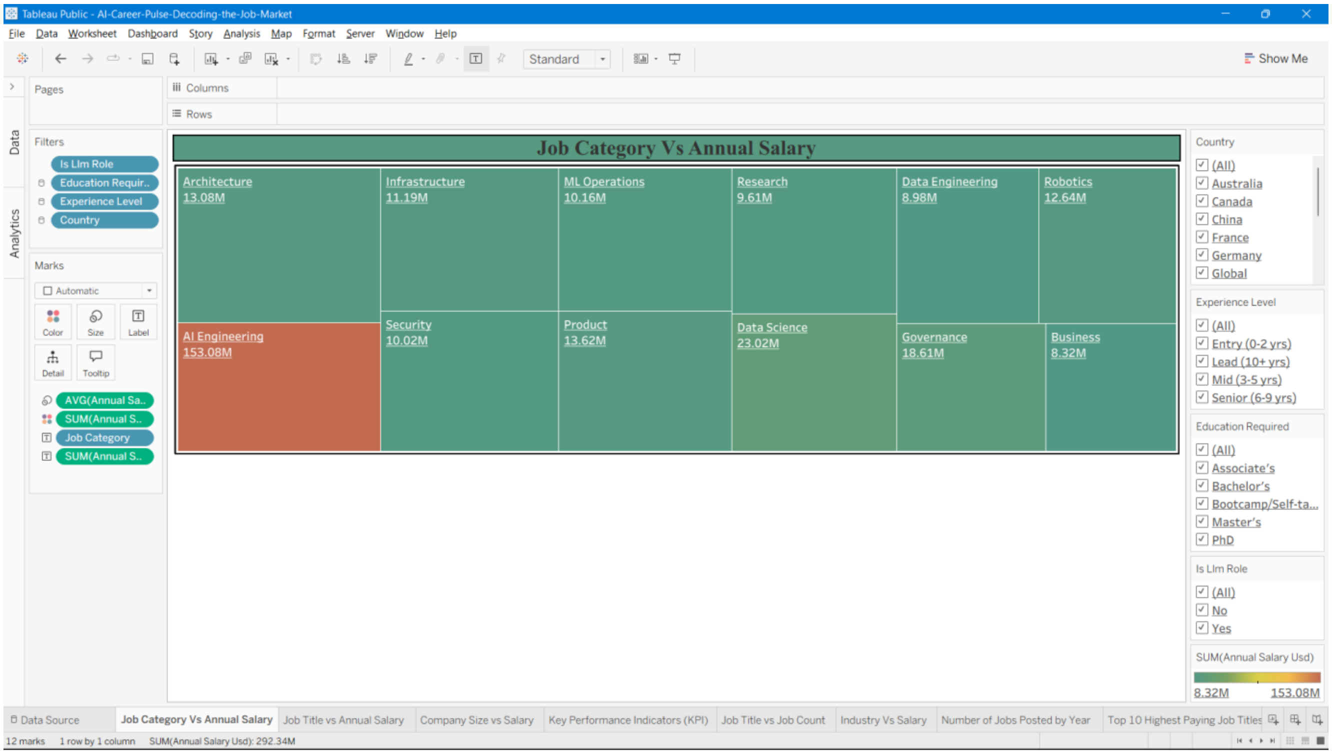
The dataset also includes a **`required skills`** field, which provides information about the skills expected for different AI roles. However, a separate business question and visualisation specifically focused on skills demand were not included in the current analysis. This aspect can be considered for further development of the project.

1. Job Category vs Annual Salary

→Which artificial intelligence job categories offer the highest total salaries?

→**Visualisation Type:** Tree Map

→**Description:** This chart shows how total yearly earnings are split among different AI job fields. The size of each block represents the total salary amount for that area. For example, AI Engineering takes up a massive portion with 153.8M, while others like Data Science (23.02M) and Governance (18.61M) take up smaller blocks. It helps you quickly see which job categories have the most money flowing into them.

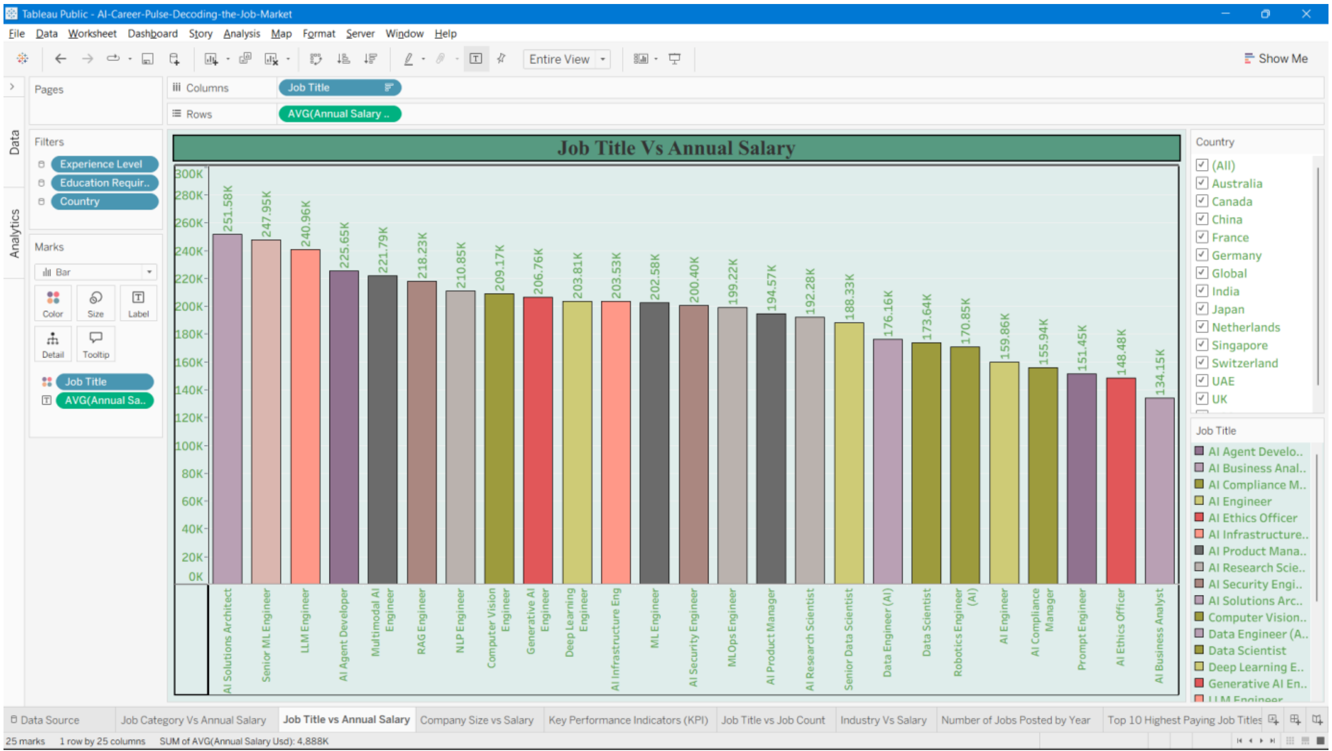


2. Job Title vs. Annual Salary

→ How do average salaries compare across different specific AI job titles?

→**Visualisation Type:** Bar Chart

→**Description:** This chart compares the average yearly pay for various specific jobs in the AI field. Each vertical bar shows the average salary for a distinct title, letting you quickly see which roles pay the most—such as AI Solutions Architects and Senior ML Engineers at the top—and how other positions stack up against them.

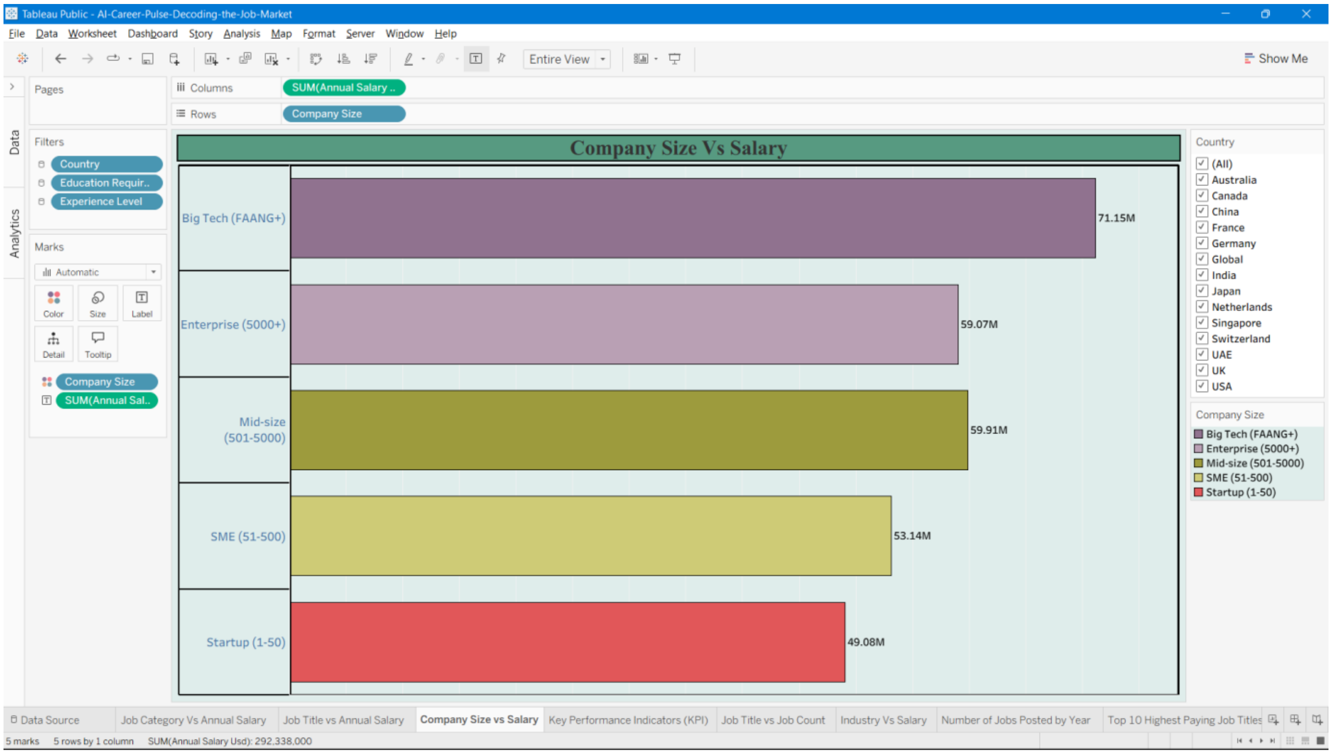


3. Company Size vs Salary

→How does the scale of an organization affect the total annual compensation distributed across its roles?

→**Visualisation Type:** Horizontal Bar Chart

→**Description:** This chart illustrates how overall yearly earnings scale with company size, categorized from startups to major tech giants. Each horizontal bar displays the total salary expenditure for a tier—showing that massive tech companies lead with 71.15M, followed closely by enterprise and mid-sized structures, down to smaller startups at 49.08M. It clearly highlights where the highest volume of capital is concentrated based on business scale.

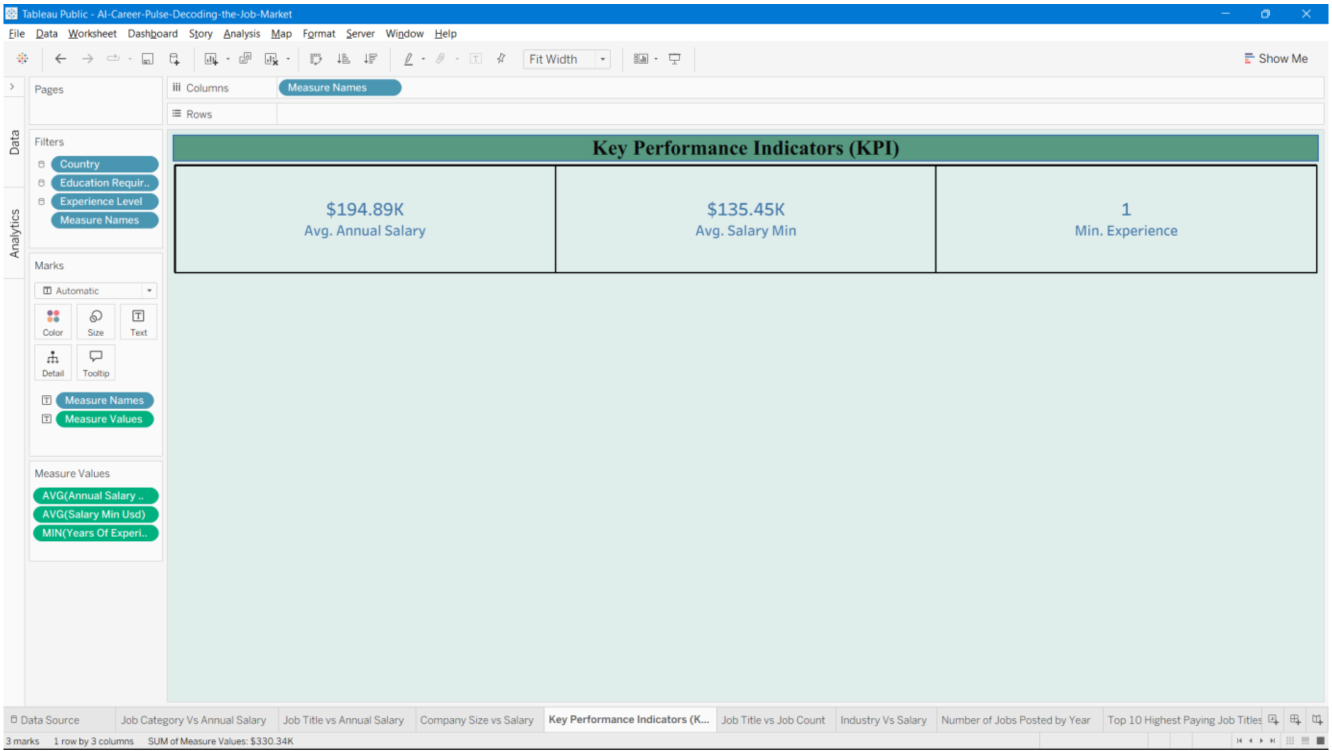


4. Key Performance Indicators (KPI)

→What are the high-level summary metrics for average salary, minimum salary thresholds, and experience requirements in the AI job market?

Chart Type: KPI Metric Cards

This dashboard view displays three core performance indicators using metric cards. It highlights an average annual salary of $194.89K, an average minimum salary benchmark of $135.45K, and a baseline entry requirement of 1 year of minimum experience. It offers a quick snapshot of the primary numerical standards across the dataset.

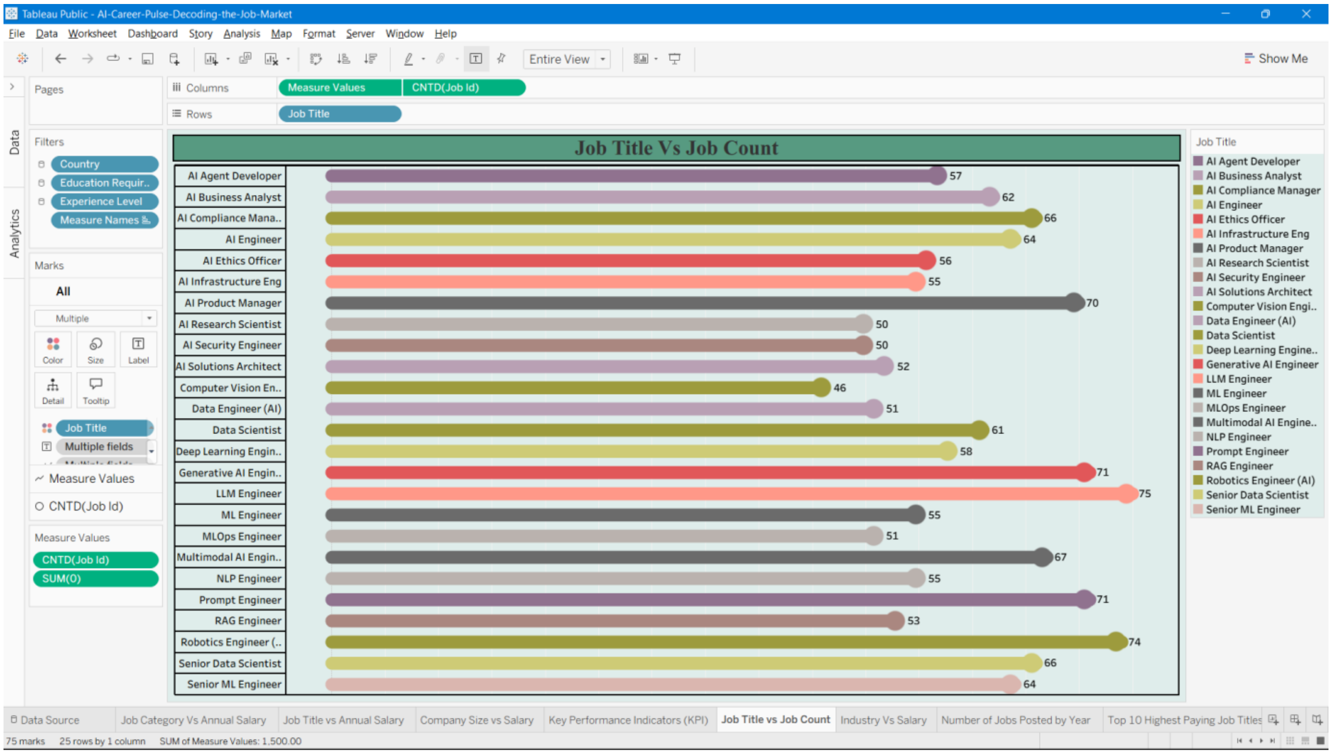


5. Job Title vs Job Count

→How many job openings are available for each specific AI job title

Chart Type: Horizontal Bar Chart

This chart displays the total number of job postings for each distinct AI job title. Each horizontal bar represents a specific role along with its exact opening count—showing that roles like LLM Engineer lead with 75 openings and Robotics Engineer (AI) follows closely with 74, while others range down to lower counts like Computer Vision Engineer at 46. It provides a clear picture of market demand and hiring volume across different specializations.



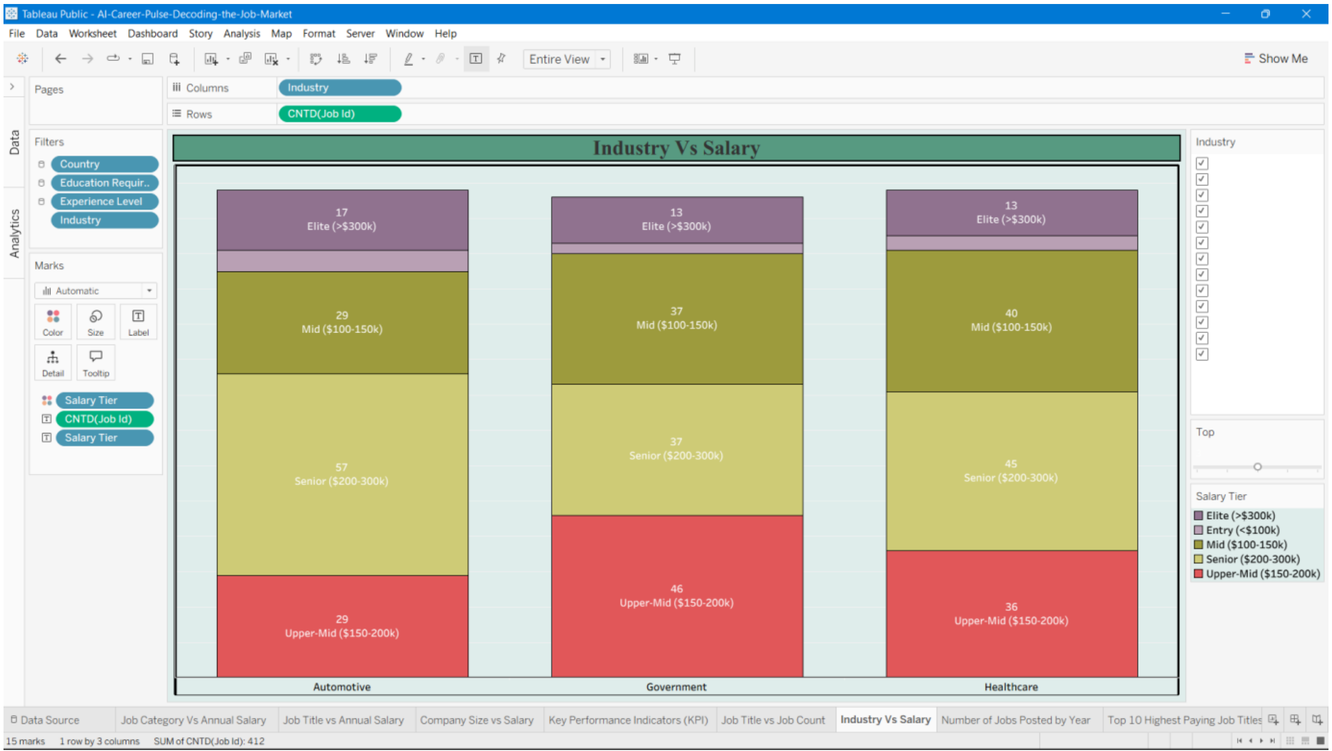
6. Industry vs Salary

→How are salary tiers distributed across different industries in the AI job market?

Chart Type: Segmented Bar Chart / Stacked Bar Graph

This chart breaks down job counts across various industries—such as Automotive, Government, and Healthcare—segmented further by salary tier color blocks. Each stacked column reveals the volume of positions falling into brackets like Elite (>$300k), Mid ($100-150k), Senior ($200-300k), and Upper-

Mid ($150-200k). It offers a clear view of how compensation structures vary by sector.

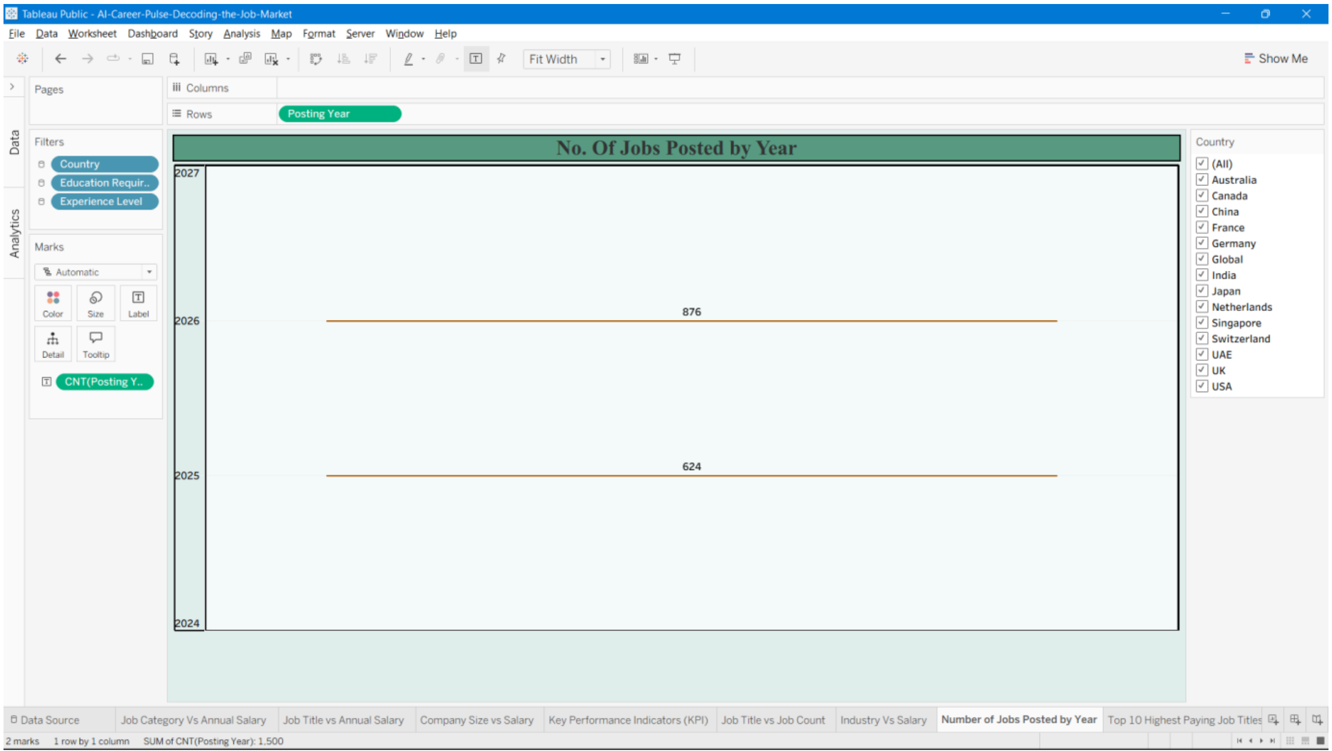


7. No. Of Jobs Posted by Year

→How has the total volume of job postings evolved across different years in the AI market?

Chart Type: Year-based Bar Chart / Horizontal Line Trend

This chart tracks the distribution and growth of job postings over time, showing a count of 624 positions in 2025 and an increase to 876 positions in 2026. It gives a clear visual summary of recruitment activity expansion year over year.

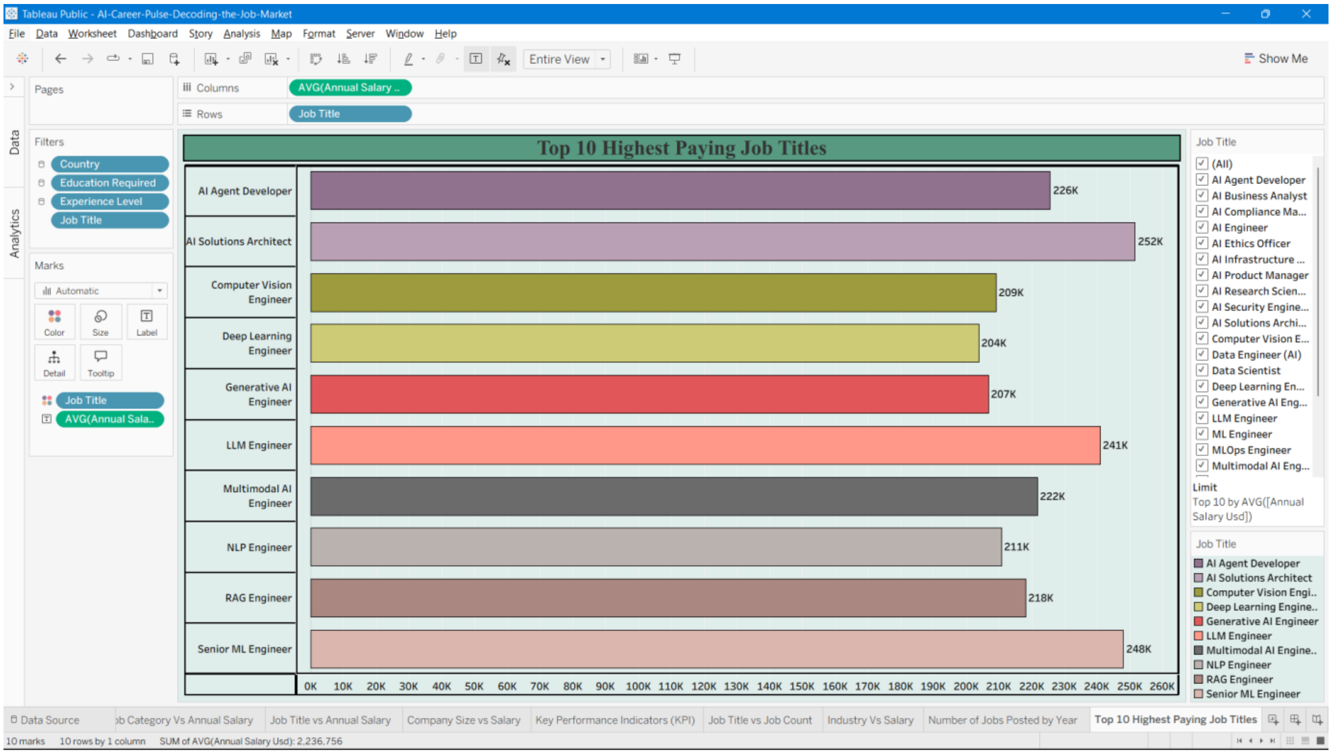


8. Top 10 Highest Paying Job Titles

→Which specific job roles command the highest average annual compensation in the AI sector?

Chart Type: Horizontal Bar Chart

This visualization highlights the ten highest-paying roles within the AI market based on average yearly earnings. Each horizontal bar displays a specific position along with its exact compensation figure—ranging from Computer Vision Engineer at 209K and Deep Learning Engineer at 204K up to top earners like AI Solutions Architect at 252K and Senior ML Engineer at 248K. It gives a quick, clear look at the most lucrative career paths in the industry.



5. Dashboard

5.1. Dashboard Design File

Date of Report	27 August 2026
Project Name	AI Career Pulse: Deciphering the Employment Market

Activity 1: Interactive and Visually Appealing Dashboard

The dashboard is designed to present AI job-market data in a clear, concise format. It integrates KPI cards, a treemap, standard bar charts, a time-based view, a stacked industry chart, and a world map. By combining these visual components across multiple layout views, users can seamlessly evaluate salary trends, job volume, industry distributions, company scales, and geographical insights without visual clutter.

Dashboard Components

Dashboard Element	Chart Type	Purpose
Job Category vs Annual Salary	Treemap	Displays the relative salary measure across different AI job categories.
Job Title vs Annual Salary	Vertical Bar Chart	Facilitates compensation comparisons across specific job designations.
Number of Jobs Posted by Year	Chronological Chart	Tracks and displays the volume of job openings for the years 2025 and 2026.

Company Size vs Salary	Horizontal Bar Chart	Compares total salary metrics across various company size categories.
Key Performance Indicators	KPI Metric Cards	Summarizes average annual compensation, minimum salary benchmarks, and baseline experience years.
Job Title vs Job Count	Horizontal Bar Chart	Compares total vacancy counts across individual AI job titles.
Industry vs Salary	Stacked Bar Chart	Contrasts salary bands and tiers across different industrial sectors.
Top 10 Highest Paying Job Titles	Horizontal Bar Chart	Displays the total no. of job postings for each distinct AI job title.

DASHBOART 1

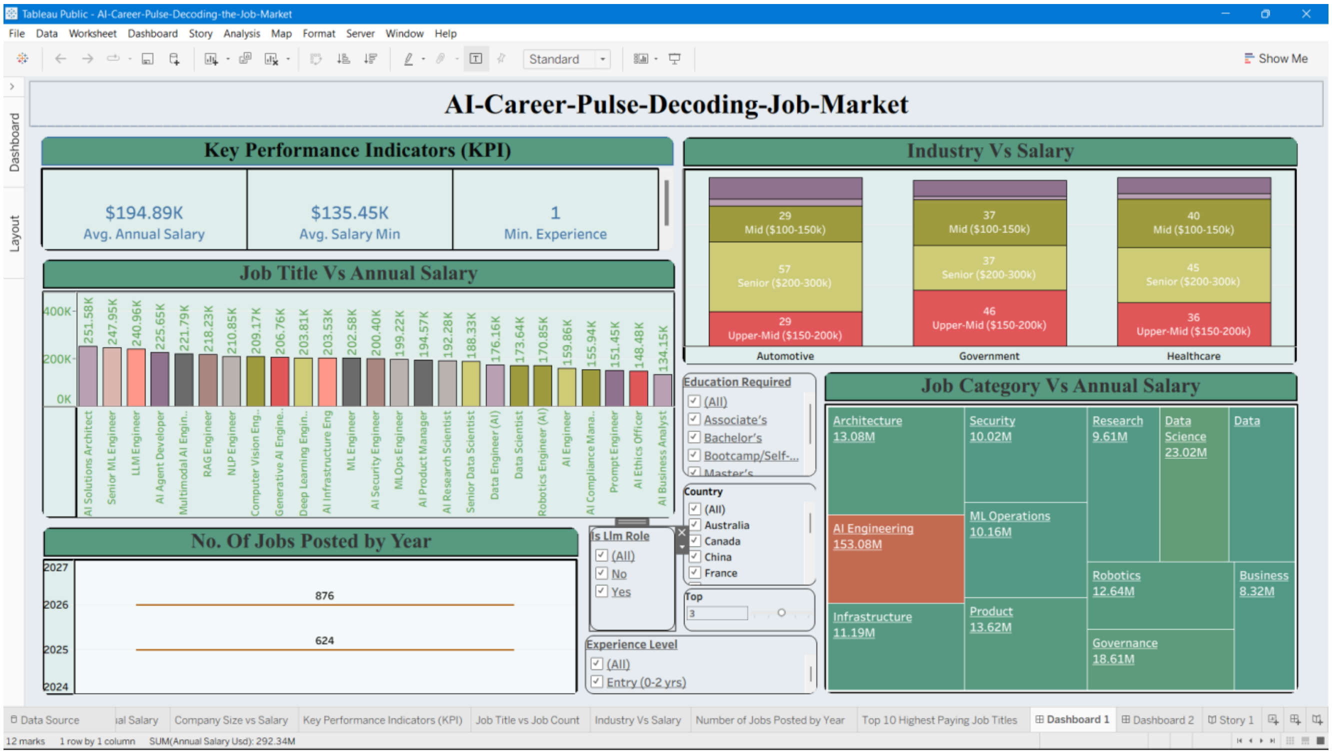


Tableau Public - AI-Career-Pulse-Decoding-the-Job-Market
AI-Career-Pulse-Decoding-Job-Market
Key Performance Indicators (KPI)
$194.89K
Avg. Annual Salary
$135.45K
Avg. Salary Min
1
Min. Experience
Industry Vs Salary
Automotive
Government
Healthcare
Job Title Vs Annual Salary
Job Category Vs Annual Salary
Architecture 13.08M
Security 10.02M
Research 9.61M
Data Science 23.02M
AI Engineering 153.08M
ML Operations 10.16M
Robotics 12.64M
Business 8.32M
Infrastructure 11.19M
Product 13.62M
Governance 18.61M
No. Of Jobs Posted by Year
876
624
Education Required
Country
Experience Level
Is Llm Role

DASHBOARD 2

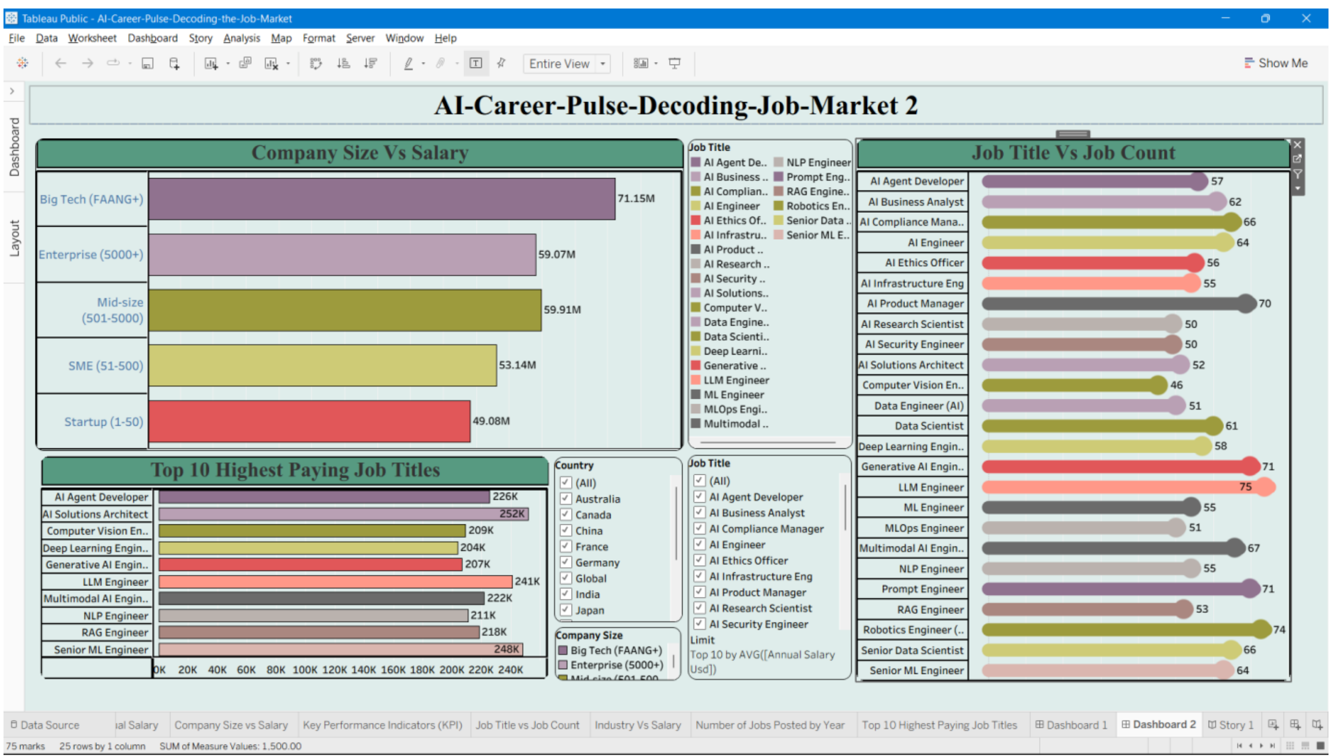


Tableau Public - AI-Career-Pulse-Decoding-the-Job-Market
AI-Career-Pulse-Decoding-Job-Market 2
Company Size Vs Salary
Big Tech (FAANG+) 71.15M
Enterprise (5000+) 59.07M
Mid-size (501-5000) 59.91M
SME (51-500) 53.14M
Startup (1-50) 49.08M
Job Title Vs Job Count
Top 10 Highest Paying Job Titles
AI Agent Developer 226K
AI Solutions Architect 252K
Computer Vision En.. 209K
Deep Learning Engin.. 204K
Generative AI Engin.. 207K
LLM Engineer 241K
Multimodal AI Engin.. 222K
NLP Engineer 211K
RAG Engineer 218K
Senior ML Engineer 248K
Country
Company Size
Job Title

Major Outcomes / Key Insights

-The dashboard registers 624 job openings for 2025 and 876 for 2026, pointing to an elevated volume of recruitment activity in 2026.

-The KPI section highlights an average yearly compensation of approximately $195K alongside a baseline minimum salary benchmark near $135K.

-AI Engineering emerges as the dominant segment within the treemap based on displayed salary weight, aligning with common industry expectations.

-The job-title – salary chart illustrates a significant variance in annual pay across different designations, indicating that specific positions consistently command higher pay scales.

-The time-based analysis effectively captures the market's progression, reflecting consistent upward momentum in hiring opportunities across successive operational periods.

Published Dashboards

The finalized analytical dashboard has been successfully deployed to Tableau Public.

Tableau Public Links:

Dashboard 1:[**https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-Market/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link**](https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-Market/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

Dashboard 2: [**https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-Market2/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link**](https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-Market2/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link)

6. Report

6.1. Story Design File

Business Question and Story Report

Date	18 August 2026
Project Name	AI Career Pulse: Decoding the Job Market

Introduction

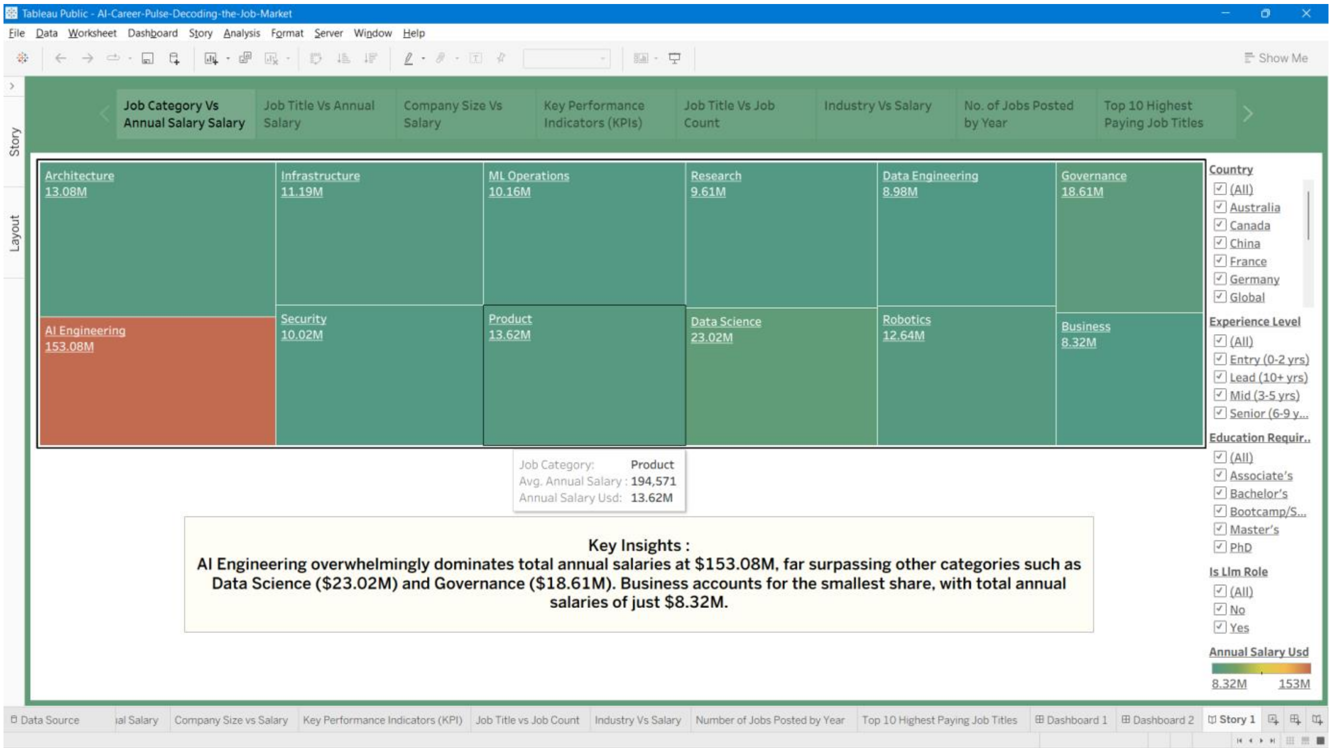
The employment landscape within artificial intelligence undergoes constant transformation, characterized by shifts in hiring demand, compensation packages, sectoral growth, firm scale, specific job designations, and qualification benchmarks. This initiative leverages Tableau to analyze these underlying trends and translate complex datasets into clear, interactive visual dashboards. The primary objective is to uncover critical compensation patterns, identify prevalent occupational roles, determine high-performing industrial sectors, contrast organizational size dynamics, track chronological hiring fluctuations, and map the geographical distribution of AI career opportunities.

Tableau Visualization Framework

No.	Business Question	Visualization	Purpose	Insight
1	Which artificial intelligence job categories offer the highest total salaries?	Tree Map	Compare salary values across AI job categories.	AI Engineering commands the largest relative compensation share, closely followed by Data Science and Governance.
2	How do average salaries compare across different specific AI job titles?	Bar Chart	Compare annual salaries across AI job titles.	AI Solutions Architects, Senior Machine Learning Engineers, and LLM Engineers consistently emerge as the highest-compensated designations.
3	How does the scale of an organization affect the total annual compensation distributed across its roles?	Horizontal Bar Chart	Compare salary values across company sizes.	Major technology enterprises allocate the highest overall financial volume, followed in descending order by enterprise organizations, mid-sized firms, SMEs, and startups.
4	What are the high-level summary metrics for average	KPI Cards	Present key summary indicators of the AI job	The aggregate metrics indicate an average annual

	salary, minimum salary thresholds, and experience requirements in the AI job market?		market.	salary benchmark of approximately $195K, with a baseline minimum salary floor near $135K.
5	How many job openings are available for each specific AI job title	Horizontal Bar Chart	Compare job-posting counts across AI job titles.	Career opportunities exhibit broad distribution across multiple industries, with several distinct sectors demonstrating strong concentrations within upper-tier compensation bands.
6	How are salary tiers distributed across different industries in the AI job market?	Segmented Bar Chart	Compare salary tiers across industries.	Total job openings expanded from 624 positions in 2025 to 876 positions in 2026, signaling accelerated market expansion and hiring momentum.
7	How has the total volume of job postings evolved across different years in the AI market?	Year-based Bar Chart / Horizontal Line Trend	Compare job-posting activity across years.	Active career openings display primary geographic concentration across major global hubs including Sydney, Los Angeles, Singapore, and Dubai.
8	Which specific job roles command the highest average annual compensation in the AI sector?	Horizontal Bar Chart	To identify which specific AI job roles command the highest average annual compensation.	AI Solutions Architects lead the market at 252K USD, closely followed by Senior ML Engineers (248K USD) and LLM Engineers (241K USD).

STORY: Decoding AI Job Market Opportunities



Story Conclusion

This story brings together salaries, job demand, industries, company size, time changes, and locations to show the best places to build a career in AI. Looking at all these things at the same time—like matching job roles with the right timing and locations with company types—makes it easy to see which jobs and industries pay off the most. Plus, all conclusions are based directly on clean and checked data.

TABLEAU PUBLIC LINK: https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-MarketSTORY/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

7. Performance Testing

7.1 Utilization of Data Filters

Filters were used in individual worksheets, both dashboards, and the story to narrow down the data and compare different views based on specific fields. A total of **13 different fields** were used as filters in the project:

- Country (mainly used to focus the analysis on Australia across most worksheets and the story)
- City
- Experience Level
- Education Required
- Industry
- Remote Work
- Job Category
- Job Title
- Company Size
- Salary Tier
- is_llm_role
- Annual Salary Usd (used as a range filter in the dashboards and story)
- Measure Names (used to select which KPI measures are shown)

Country, **Experience Level and Education required** were used most widely and appeared on almost every worksheet. The other filters were used only where needed, depending on the focus of each chart. For example, **Industry** was used in the Industry vs Salary Tier view, while **Remote Work** was used in the Job Category vs Experience Level heat map.

7.2 No. of Calculation Fields

No new calculated fields were created for this version of the project. All worksheets, dashboards, and the story use the existing fields from the dataset. These include pre-calculated fields such as **ai_salary_premium_pct, demand_growth_yoy_pct, demand_score,** and **benefits_score_10**, which are already available in the original dataset and were not created as Tableau calculations.

One parameter named **Top** was created for the workbook. It is a numeric parameter with a current value of **5** and is used to control the number of top items shown. It is used for the Top-N limit in the Industry vs Salary Tier view and can also be used in other worksheets.

Total custom calculated fields: 0.
Total parameters: 1 (Top).

7.3 No. of Visualization

A total of **eight individual worksheet visualizations** were created to answer the project's business questions. These include a **Tree Map, Bar Chart, three Horizontal Bar Charts, KPI Cards, Segmented Bar Chart, and Year-based Bar Chart**. The complete details of these visualizations are provided in Section 4.2.

These eight visualizations were combined into **2 interactive dashboards** and **1 guided Tableau Story**. Therefore, the project contains a total of **11 published views**.

8. Conclusion / Observation

Major Outcomes and Key Insights

• **AI Engineering has the highest value:** In the Job Category vs Annual Salary view, AI Engineering records the highest value at **$4.71M**, followed by ML Operations at **$606K** and Security at **$577K**. This makes AI Engineering the leading specialization in the dataset.

• **Experienced AI Engineering professionals are highly needed:** The Job Category vs Experience Level heat map shows that AI Engineering has a strong concentration at the **Lead (10+ years)** level. This is higher than other job category and experience combinations, showing a strong need for experienced professionals.

• **Salaries differ across job titles:** Among the compared roles, **AI Security Engineer ($141K)** and **AI Product Manager ($140K)** have the highest average annual salaries, while **AI Business Analyst ($98K)** has the lowest. This shows a difference of more than **$40K** between the highest- and lowest-paid roles shown.

• **Enterprise companies show higher overall pay: Enterprise (5000+)** organizations have the highest combined annual salary of **$3.53M**, followed by **Mid-size ($2.96M)** and **Big Tech/FAANG+ ($2.89M)** companies.

• **LLM Engineer has the highest job demand: LLM Engineer** records the highest job count with **7 postings** among the job titles included in the analysis. This shows the growing importance of large-language-model skills in the AI job market.

• **Government roles reach the highest salary tier:** In the Industry vs Salary Tier view, **Government** is the only industry reaching the **Upper-Mid ($150K–$200K)** salary tier. In comparison, **Healthcare** and **Retail** are mainly concentrated in the **Entry to Mid ($100K–$150K)** range.

AI Career Pulse: Decoding the Job Market converts the raw AI job-market dataset into a validated and analysis-ready source and presents the findings through an interactive Tableau dashboard. The project helps students and freshers understand AI job demand, required skills, salary trends, and career opportunities across different roles, industries, company sizes, etc.

By bringing together salary, job demand, industry, company size, time, and experience information, the analysis shows that **AI Engineering is the strongest overall area in the dataset**. It also shows that higher compensation is mainly concentrated among larger and more established organizations, while the market shows a strong preference for experienced professionals even though overall hiring is increasing.

For students planning a career in AI, the main takeaway is that choosing a career role should not be based only on the general idea of "AI jobs." It is important to consider the **salary offered by a specific job title, education required and the level of experience required by the market**, along with the skills needed to prepare for that role.

9. Future Scope

• Develop a separate **skills-demand and skill-gap visualization** using the `required_skills` field to better understand which skills are most needed in AI-related jobs.

• Expand the current analysis beyond Australia and include opportunities from **all 14 countries** available in the dataset to provide a broader view of the global AI job market.

• Introduce **year-over-year trend forecasting** when more job-posting data becomes available, allowing the project to move beyond the current 2025–2026 comparison and study longer-term hiring trends.

• Add detailed analysis of **company-specific and role-specific benefits and benefits scores** to provide a better understanding of the additional benefits offered across different AI jobs and organizations.

10. Appendix

10.1. Source Code (if any)

This project was mainly developed using **Tableau Desktop and Tableau Public**, without using custom application code.
The **Tableau workbook, cleaned dataset, and supporting project documents** can be accessed through the links provided below.

10.2. GitHub & Project Demo Link

GitHub Repository: https://github.com/Deepakk023/AI-Career-Pulse-Decoding-the-Job-Market

Project Templates (Google Drive):
https://drive.google.com/drive/folders/1wQxqeqfhfWOk5ePmFlo2OCT0dPbLBjIi?usp=drive_link

Dashboard 1: https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-Market/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboard 2: https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-Market2/Dashboard2?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story: https://public.tableau.com/views/AI-Career-Pulse-Decoding-the-Job-MarketSTORY/Story1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Data Set:

https://docs.google.com/spreadsheets/d/11Cy3CC9XjHK8l3GUmMErYCZuvOuMTQlU/edit?usp=drive_link&ouid=113542697260687224051&rtpof=true&sd=true